CSE 271s : System Dynamics & Control Components
Lab 4





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Question 1:

a)

$$\frac{B(s)}{A(s)} = \frac{2s^3 + 5s^2 + 3s + 6}{s^3 + 6s^2 + 11s + 6}$$

```
Command Window
>> lab4

num =

     2     5     3     6


den =

     1     6    11     6


r =

   -6.0000
   -4.0000
    3.0000


p =

   -3.0000
   -2.0000
   -1.0000


k =

     2

fx
```

b)

$$\frac{B(s)}{A(s)} = \frac{s^2 + 2s + 3}{(s+1)^3} = \frac{s^2 + 2s + 3}{s^3 + 3s^2 + 3s + 1}$$

```
Editor - D:\matlab\projects\lab4.m
lab4.m
5
6
7     num=[1 2 3 ]
8
9     den=[ 1 3 3 1 ]
10
11    [r,p,k]=residue(num,den)
12
13
14    num1=[10];
15    den1=[1 2 10];
```

```
Command Window

r =

    1.0000
   -0.0000
    2.0000


p =

   -1.0000
   -1.0000
   -1.0000


k =

     []

fx
```

Question 2:

a)

$$G_1(s) = \frac{10}{s^2 + 2s + 10} = \frac{\text{num1}}{\text{den1}}, \qquad G_2(s) = \frac{5}{s + 5} = \frac{\text{num2}}{\text{den2}}$$

```
>> num1 = [10];
den1 = [1 2 10];
num2 = [5];
den2 = [1 5];
[num,den] = series(num1,den1,num2,den2);
printsys(num,den)
```

Command Window

```
num/den =

              50
   ----------------------
   s^3 + 7 s^2 + 20 s + 50

num/den =

      5 s^2 + 20 s + 100
   ----------------------
   s^3 + 7 s^2 + 20 s + 50

num =

     0     0    10    50


den =

     1     7    20   100


num/den =

           10 s + 50
   ----------------------
   s^3 + 7 s^2 + 20 s + 100
```

b)

$$\frac{Y(s)}{U(s)} = \frac{s}{(s+10)(s^2+4s+16)}$$

$$= \frac{s}{s^3+14s^2+56s+160}$$

```
>> num = [1 0];
>> den = [1 14 56 160];
>> [A,B,C,D] = tf2ss(num,den)
```

Command Window

```
A =

   -14   -56  -160
     1     0     0
     0     1     0


B =

     1
     0
     0


C =

     0     1     0


D =

     0


num =

         0         0   25.0000    5.0000
```

c)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5 & -25 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 25 \\ -120 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

```
>> A = [0 1 0; 0 0 1; -5 -25 -5];
>> B = [0; 25; -120];
>> C = [1 0 0];
>> D = [0];
>> [num,den] = ss2tf(A,B,C,D)
```

Command Window

```
num =

         0         0   25.0000    5.0000


den =

    1.0000    5.0000   25.0000    5.0000
```

d)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -25 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

```
>> A= [0 1; -25 -4];
>> B= [1 1; 0 1];
>> C= [1 0; 0 1];
>> D= [0 0; 0 0];
>> [NUM,den]= ss2tf(A,B,C,D,1)

NUM =

     0     1     4
     0     0   -25


DEN =

    1.0000    4.0000   25.0000


NUM =

          0    1.0000    5.0000
          0    1.0000  -25.0000


DEN =

    1.0000    4.0000   25.0000
```

Question 3:

Command Window

```
x =

3*t + t^2*exp(-a*t)


ans =

2/(a + s)^3 + 3/s^2


x =

sin(3*t)*exp(-t/10)


ans =

3/((s + 1/10)^2 + 9)


x =

sin(3*t)*exp(-(3*t)/10)


ans =

3/((s + 3/10)^2 + 9)
```

Command Window

```
x =

sin(3*t)*exp(t/10)


ans =

3/((s - 1/10)^2 + 9)


x =

sin(3*t)*exp((3*t)/10)


ans =

3/((s - 3/10)^2 + 9)


x =

sin(3*t)*exp(t/10)


x4 =

sin(3*t)*exp(t/10)
```

f)

```
>> fplot(x2,[0 10])
>> hold on
>> fplot(x3,[0 10])
>> hold on
>>
```

